Luminary Journal

Product Requirements & Technical Specification

Version 1.0 — Self-Directed Tech Exploration

1. Project Overview

Luminary Journal is a web-based skincare routine analyzer and personal skin tracking application. It enables users to build a detailed weekly skincare routine, receive AI-powered ingredient analysis and conflict detection, track their skin health over time through weekly check-ins, and simulate how routine changes might affect their skin going forward.

The core differentiator from existing tools like Skinsort is the combination of a structured weekly routine builder, personalized predictive modeling based on logged history, and a swap simulator that shows projected vs. actual skin response over time. The aesthetic direction is soft, editorial, and beauty-forward — clean and cute with a professional feel.

1.1 Project Goals

- Build a functional web prototype within approximately one week
- Provide genuinely useful ingredient-level analysis tailored to the user's skin profile
- Make weekly skin tracking feel effortless and habit-forming
- Differentiate from competitors through predictive swap simulation and projected vs. actual tracking
- Produce a portfolio-quality project suitable for recruiter presentation

1.2 Target User

Skincare-interested individuals (primarily ages 18–35) who maintain a multi-step routine and want to understand how their products interact with each other and with their skin type. Users range from casual skincare enthusiasts to more dedicated "ingredient-aware" consumers familiar with terms like actives, barrier support, and comedogenicity.

2. Technology Stack

Layer	Technology	Purpose
Frontend	React	Single-page application, component-based UI
Auth	Supabase Auth + Google OAuth	Social login via Google, session management
Database	Supabase (PostgreSQL)	User profiles, routines, logs, custom products
Storage	Supabase Storage	Any future media/asset needs
AI Analysis	Claude API (claude-sonnet-4-20250514)	Ingredient analysis, scoring, explanations, projections
Serverless Functions	Vercel Edge Functions	Proxy Claude API calls securely
Hosting	Vercel	Deployment, CI/CD from GitHub
Product Database	Open Beauty Facts + curated dataset	Product search and ingredient data

2.1 Key Technical Notes

- Claude API calls must be proxied through Vercel Edge Functions — never expose API keys on the client side
- Supabase Row Level Security (RLS) must be enabled so users can only access their own data
- All AI analysis calls should be debounced (minimum 2 second delay after last change) to avoid excessive API usage
- The app is web-only for the prototype. Mobile responsiveness is a nice-to-have but not required for v1

3. Navigation Structure

The app uses a top navigation bar with five primary sections. Navigation is only accessible after the user has completed onboarding.

Nav Item	Description
Dashboard	Overview of skin trends, weekly history chart, recent log entries

Nav Item	Description
Routine	Weekly calendar routine builder + live analyzer output
Simulate	Swap simulator for projecting routine changes
Journal	Weekly check-in flow and log history
Profile	Skin profile, medication checklist, account settings

4. Onboarding Flow

New users are guided through a linear three-step onboarding sequence before reaching the main app. Each step is a distinct screen with a progress indicator.

Step 1: Account Creation

- Welcome screen with app name, tagline, and brief value proposition
- Single “Continue with Google” button for social login via Supabase Auth
- No email/password option in v1

Step 2: Skin Profile Setup

User fills in their skin profile. All fields are required except sensitivity level notes.

Field	Input Type	Options / Notes
Skin Type	Single select dropdown	Normal, Dry, Oily, Combination, Sensitive
Skin Concerns	Multi-select tags	Acne, Blackheads, Dryness, Redness, Hyperpigmentation, Fine Lines, Dullness, Large Pores, Texture, Dark Circles
Sensitivity Level	Slider 1–5	1 = Not sensitive, 5 = Very reactive
Medications	Multi-select checklist	See Section 4.1 for full list

4.1 Medication Checklist Options

Display as a checklist with a brief one-line note on each explaining skin relevance:

- Isotretinoin / Accutane — extreme dryness, sun sensitivity, no exfoliants
- Oral Retinoids — similar to Accutane, retinoid sensitivity
- Tretinoin / Tazarotene (topical prescription) — retinoid in routine
- Spironolactone — hormonal, affects oil production
- Hormonal Birth Control — may affect acne and oiliness
- Doxycycline / Minocycline — antibiotic, photosensitivity risk
- Methotrexate — skin fragility, avoid strong actives
- Corticosteroids (oral or topical) — barrier thinning
- Lithium — can trigger or worsen acne
- None of the above

Step 3: First Routine Build

User builds their first week's routine using the calendar interface (see Section 6 for full Routine Builder spec). After completing and confirming the calendar, a brief "Initial Analysis" screen appears before redirecting to the Dashboard.

4.2 Initial Analysis Screen

Shown once after first routine build. Displays:

- A summary of how the user's product set relates to their skin type and concerns
- Any active conflicts or high-risk combinations detected in the routine
- A "View Full Analysis" CTA that leads to the Routine section

This screen is intentionally brief — 2–3 sentences of AI-generated insight plus flagged conflicts only. No scores shown here.

5. Product Database

5.1 Data Sources

- Primary: Curated dataset of 500–800 commonly used products across popular brands (CeraVe, La Roche-Posay, The Ordinary, Neutrogena, Drunk Elephant, Paula's Choice, Cetaphil, Tatcha, Cosrx, Bioderma, etc.)
- Secondary: Open Beauty Facts open-source database (https://world.openbeautyfacts.org) — free, structured, includes INCI ingredient lists
- Tertiary: User-submitted custom products saved to their personal profile

5.2 Product Data Schema

Field	Type	Notes
id	UUID	Primary key
name	String	Product name e.g. “CeraVe Moisturizing Cream”
brand	String	Brand name
category	String	Cleanser, Toner, Serum, Moisturizer, SPF, Treatment, Mask, Eye Cream, Oil, Other
ingredients	Text / String[]	Full INCI ingredient list
is_custom	Boolean	True if user-created
created_by	UUID (user_id)	Null for global products, user id for custom

5.3 Custom Product Creation

If a user cannot find their product in the search results, they can create a custom product:

- Enter a custom name (e.g. “That Trader Joe’s Toner”)
- Paste in the full INCI ingredient list from the product packaging
- Select product category
- Save to their personal product library

Custom products are saved permanently to the user’s profile and appear in all future product searches, routine building, and the swap simulator just like database products.

6. Routine Builder

Accessed via the “Routine” nav item. The screen is split into two main areas: the weekly calendar on the left and the analyzer output panel on the right.

6.1 Calendar Interface

A 7-day grid with AM and PM rows for each day, giving 14 total slots. Each slot displays the products assigned to it as small product pills/tags.

Slot Interaction

- Clicking any slot (e.g. “Monday PM”) highlights it with a soft accent color
- A sidebar panel slides in from the right with a product search bar
- User types to search the product database or scrolls their saved products
- Selecting a product adds it to the highlighted slot
- User can click another slot directly to switch focus without closing the sidebar
- A “Done” or close button confirms and collapses the sidebar

Copy / Fill Shortcuts

- “Apply to all mornings” — copies current AM slot’s products to all 7 AM slots
- “Apply to all evenings” — same for PM
- “My routine is the same every day” — fills all 14 slots with the same product set
- User can then edit individual slots to add exceptions (e.g. exfoliant only on Tuesday/Friday)

Product Order Within Slots

- Products within each slot are orderable by clicking and dragging within that slot
- A subtle numbered badge shows the application order (1, 2, 3...)
- A soft non-intrusive tip is shown once: “Tip: apply products thinnest to thickest consistency”
- No automatic flagging of order as “wrong” — the tip is informational only

Week Persistence

- At the start of each new week, the previous week’s routine is pre-loaded into the calendar
- User only needs to make changes if something is different
- This also occurs during the weekly Journal check-in flow (see Section 8)

6.2 Analyzer Output Panel

Displayed to the right of the calendar. Re-runs automatically when the routine changes, with a 2-second debounce. Previous analysis is replaced each time.

Per-Product Scores

Each product in the routine receives five scores on a 1–10 scale:

Score	What It Measures
Irritation Risk	Likelihood of causing redness, stinging, or sensitivity reactions based on ingredients and user’s sensitivity level
Redundancy	Degree to which this product’s functions are already covered by other products in the routine

Score	What It Measures
Barrier Support	How well the product supports or restores the skin moisture barrier
Comedogenicity	Pore-clogging risk based on known comedogenic ingredient ratings
pH Compatibility	How well this product's pH range works when layered with adjacent products in the routine

Routine-Level Output

- Flagged conflicts: specific ingredient combination warnings (e.g. high-percentage AHA + retinol)
- Duplicate function alerts: products performing the same role
- Pore-clogging risk products: individual products with high comedogenicity scores
- AI-generated explanation: 3–5 sentences summarizing the overall routine health, key risks, and recommendations, personalized to the user's skin profile

7. Simulate (Swap Simulator)

Accessible via the "Simulate" nav item. Also reachable via a "Simulate a swap" button in the Routine builder that navigates here with the current routine preloaded.

7.1 Interface

- Displays the user's current routine
- User can select a product to remove from the routine
- User can search for a product to add (same search sidebar as Routine builder)
- Both actions can be combined: remove X and add Y simultaneously
- A "Project this change" button runs the simulation

7.2 Projected Output

- Re-runs all five scores for the modified routine
- Shows a score comparison table: current routine scores vs. projected scores
- AI generates a 3–5 sentence projection narrative: expected adjustment period, likely improvements, potential risks
- The projected change is also stored silently in the database as a pending projection for the affected weeks

7.3 Projected vs. Actual Tracking

This is a passive background feature that runs regardless of whether the user actively uses the Simulate section:

- Every time a routine change is logged during the weekly Journal check-in, the system automatically generates a background projection for the coming weeks
- As real Journal check-ins come in after the change, actual skin metric scores are compared against the projection
- This projected vs. actual comparison is surfaced on the Dashboard chart (see Section 9)
- For the first 6 weeks of use, projections are based on general AI analysis of the routine change
- After 6+ weeks of logged data, projections begin to incorporate the user's personal historical patterns

8. Journal (Weekly Check-In)

Accessible via the "Journal" nav item. The Journal has two views: the check-in flow and the log history.

8.1 Weekly Check-In Flow

A guided multi-step flow the user completes once per week. Estimated time: 2–3 minutes.

Step 1: Routine Review

- Displays last week's calendar pre-filled
- User confirms if the routine stayed the same, or makes edits to reflect any changes
- Any changes made here are saved and trigger a background projection

Step 2: Skin Metrics

Five sliders, each 1–5, with a small illustrative icon next to each:

- Oiliness — shine/droplet icon
- Dryness — cracked texture icon
- Irritation — red flush icon
- Breakouts — small blemish icon
- Overall Glow — soft radiance icon

Step 3: Diet Tags

Quick-tap multi-select tags. User selects anything that applied this week:

- High sugar, Dairy, Alcohol, Lots of water, Low water, Processed food, Mostly whole foods, High stress, Poor sleep, Travel / environment change, Hormonal week

Completion

- A brief confirmation screen with a summary of this week's scores
- Redirects to the Dashboard on dismiss

8.2 Log History

Below the check-in prompt, a scrollable list of past weekly entry cards. Each card displays:

- Week date range
- Mini summary of the five metric scores (small colored dots or mini bar)
- Any routine changes that week (product added/removed)
- Click to expand: opens a full view of that week's log including all scores, diet tags, and a snapshot of the exact routine that week

9. Dashboard

The landing screen after onboarding. Shows a high-level overview of skin health trends and recent history.

9.1 Trend Chart

- Line chart showing all five skin metrics (oiliness, dryness, irritation, breakouts, glow) over time
- X-axis: weeks. Y-axis: 1–5 scale
- Routine change events are marked as small dot/flag markers on the timeline
- After a routine change, a dashed projected line extends forward from the change point
- As real check-ins come in, actual lines replace the projected dashed lines
- Both projected and actual lines are visible simultaneously for a period to enable visual comparison

9.2 Recent Entries

- Row of the most recent 3–4 weekly entry cards (same format as Journal log history)
- Each card is clickable to open the full week detail view

9.3 Quick Stats

- Current routine product count
- Weeks logged
- Most improved metric (based on trend over last 4 weeks)

10. Profile

Accessible via the “Profile” nav item.

- View and edit all skin profile fields (skin type, concerns, sensitivity, medications)
- View saved custom products with option to edit or delete
- Account info: Google account name/email, option to sign out
- No password management needed (Google OAuth handles auth)

11. Database Schema (Supabase / PostgreSQL)

users

Column	Type	Notes
id	UUID	Primary key, from Supabase Auth
email	String	From Google OAuth
skin_type	String	normal / dry / oily / combination / sensitive
skin_concerns	String[]	Array of concern tags
sensitivity_level	Integer (1–5)	
medications	String[]	Array of selected medication keys
created_at	Timestamp	

products

Column	Type	Notes
id	UUID	Primary key
name	String	
brand	String	
category	String	
ingredients	Text	Full INCI string

Column	Type	Notes
is_custom	Boolean	
created_by	UUID	Null for global, user_id for custom

routines

Column	Type	Notes
id	UUID	Primary key
user_id	UUID	Foreign key to users
week_start	Date	Monday of the week this routine applies to
slots	JSONB	Map of slot keys (e.g. “mon_am”) to ordered product_id arrays
created_at	Timestamp	

journal_entries

Column	Type	Notes
id	UUID	Primary key
user_id	UUID	Foreign key to users
week_start	Date	Monday of the logged week
oiliness	Integer (1–5)	
dryness	Integer (1–5)	
irritation	Integer (1–5)	
breakouts	Integer (1–5)	
glow	Integer (1–5)	
diet_tags	String[]	Selected diet tags
routine_id	UUID	The routine in effect this week
created_at	Timestamp	

projections

Column	Type	Notes
id	UUID	Primary key
user_id	UUID	Foreign key to users
triggered_by_routine_id	UUID	The routine change that triggered this projection
week_start	Date	Week this projection applies to
projected_oiliness	Float	
projected_dryness	Float	
projected_irritation	Float	
projected_breakouts	Float	
projected_glow	Float	
narrative	Text	AI-generated projection text
created_at	Timestamp	

12. AI Integration (Claude API)

12.1 Analysis Calls

Every analysis call should pass the following context to Claude:

- User's skin type, concerns, sensitivity level, and active medications
- Full product list with complete ingredient lists for each product
- The routine structure (which products are used AM vs. PM)

Claude returns a structured JSON response containing:

- Per-product scores (irritation risk, redundancy, barrier support, comedogenicity, pH compatibility) on 1–10 scale
- List of flagged conflicts with descriptions
- List of redundancy pairs
- 3–5 sentence routine summary explanation

12.2 Projection Calls

Triggered when a routine change is logged. Claude receives:

- Current and new routine comparison
- User's skin profile
- Historical journal entries if 6+ weeks are available

Claude returns:

- Projected metric scores for the next 4–6 weeks
- 3–5 sentence projection narrative

12.3 Cost Management

- Debounce all analysis calls — minimum 2 seconds after last routine edit
- Cache the last analysis result and only re-run if the routine actually changed
- Projection calls run once per routine change, not on every page load
- Use claude-sonnet-4-20250514 for all calls (best balance of quality and cost)

12.4 Future Optimization (Post-Prototype)

- Layer in a hardcoded rules engine for well-known interactions (e.g. AHA + retinol conflict, known comedogenic ingredients list) to reduce API call frequency and improve consistency
- Use AI only for nuanced analysis that rules cannot cover

13. Design & Aesthetic Direction

The visual design should feel like a soft beauty editorial — clean, cute, and intentional. Reference: beauty magazine layouts, glass skin editorial photography, pastel skincare flat lays.

13.1 Visual Direction

- Color palette: warm creams, blush pinks, soft terracottas, sage greens, off-whites
- Typography: pair a delicate serif or refined display font for headings with a clean sans-serif for body/UI text
- Illustration style: soft watercolor-adjacent spot illustrations — water droplets, fruit slices, skincare bottles, botanicals — used as accents not decorations
- UI components: rounded corners, generous whitespace, soft shadows, subtle gradients
- Avoid: clinical/medical aesthetics, harsh blacks, neon colors, anything that looks like a health app

13.2 Key UI Moments

- Onboarding: full-screen soft illustrated welcome with the Luminary Journal wordmark
- Routine calendar: clean grid with soft slot highlight states
- Analyzer scores: elegant score display (consider circular progress rings or soft bar fills)
- Dashboard chart: soft multi-line chart with the projected dashed line in a lighter tint
- Journal check-in: large, tactile sliders with illustrated icons

14. Prototype Scope & Priorities

Priority	Feature	Notes
P0 — Must Have	Google login + Supabase auth	Core infrastructure
P0 — Must Have	Skin profile setup	Required for personalized analysis
P0 — Must Have	Routine calendar builder	Core feature
P0 — Must Have	Product search + custom product creation	Required for routine builder
P0 — Must Have	AI routine analyzer with 5 scores	Core differentiator
P0 — Must Have	Weekly Journal check-in	Core tracking feature
P0 — Must Have	Dashboard with trend chart	Core output
P1 — Should Have	Swap simulator with projected scores	Key differentiator
P1 — Should Have	Projected vs. actual chart overlay	Key differentiator
P1 — Should Have	Week-over-week routine persistence	UX quality of life
P1 — Should Have	Copy to all mornings/evenings shortcut	UX quality of life
P2 — Nice to Have	Hardcoded rules engine layer	Post-prototype optimization
P2 — Nice to Have	Personalized projections after 6 weeks	Requires data accumulation
P2 — Nice to Have	Mobile responsive design	Post-prototype
P3 — Future	Native mobile app	Out of scope for v1

15. Open Questions for Development

- What is the exact JSON schema Claude should return for analysis? This should be defined and tested before building the frontend analyzer component.
- How will the Open Beauty Facts data be imported and structured in Supabase? Will it be a one-time import script or a live API call?
- What is the fallback behavior if Claude API is unavailable or returns an error during analysis?
- Should custom products created by one user ever be shared to the global product database (with permission)? Not in scope for v1 but worth deciding early for schema design.